Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/27/2006
PI Name:	Zinchenko, Alexander Ph.D.		
Project Title:	Multiphase Flow of Emulsions Through Porous Media		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Boulder	**State:**	CO
Zip Code:	80309	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
Start Date:	01/01/2005	**End Date:**	05/31/2007
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 05/31/2007 per PI (12/06) NOTE: Added to Task Book in 10/2006 per GRC		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Davis , Robert H. (University of Colorado at Boulder)		
Grant/Contract No.:	NNC05GA55G		
Performance Goal No.:			
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Task Description:	Multiphase emulsion and bubbly flows through packed beds and other porous media are important in space-based applications such as liquid and solid waste recycling and food and fuel processing. Dynamic simulation capabilities will be established to describe multiphase flows of concentrated emulsions through porous media and packed beds in the presence and absence of gravitational effects, with supporting experiments on selected test problems. The specific tasks include: (i) Development of new simulation methods for deformable drops and rigid particles - Highly efficient computer algorithms based on hybrid boundary-integral-multipole techniques will be developed to simulate flows of emulsions of deformable drops (or bubbles) through porous media and fixed beds of rigid particles with prescribed microstructure. (ii) Development of new simulation methods for spherical drops and rigid particles - For small drops or bubbles that exhibit little deformation, a purely multipole algorithm will be constructed using geometry perturbation to handle near-contact interactions and lubrication forces. Conditions when the drops or bubbles become trapped within the bottlenecks of the porous medium will be identified. (iii) Development of new methods for predicting the microstructure of packed beds - Using a novel algorithm based on random placements and mechanical equilibrium, random close and loose packings of sperical particles will be generated and used in hydrodynamical simulations of multiphase flows through these packed beds. (iv) Performance of ground-based test experiments - A focused set of microphysical experiments on single-drop interactions with one, two, or three solid particles will be performed to test and guide some of the most sensitive features of the simulations. The effects of gravity will be assessed by varying the ratio of drop and matrix densities.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Following our recent study of a 3D deformable drop squeezing through tight free-space clusters of two or three particles, we have incorporated the boundary-integral tools into a large, multipole-accelerated code. This code now allows for dynamical simulations for an arbitrary number of 3D drops owing through a periodic box containing an arbitrary number of rigidly-held solid particles under a specified average pressure gradient. A severe numerical difficulty of these squeezing simulations is a necessary very high resolution thus limiting the number of drops and particles. We made most calculations for a periodic emulsion owing through a dense cubic lattice of solid spheres ; the drop non-deformed diameter is up to two times bigger than the inner constriction diameter. Less deformable drops (Ca = 0.772) move slower through the constrictions, but they \shoot out" faster because of more surface energy accumulated. The velocity of the whole mixture ("emulsion velocity") shows the same trends. The minumum drop velocity, as a function of the capillary mumber, exprapolated to zero at a well defined Ca thus determining the critical threshold for squeezing to occur. The nature of the trapped state, though, for subritical Ca, is unclear and is being investigated. Depending on the outcome, the van der Waals forces or other short-ranged effects may or may not come into play during squeezing. For supercritical Ca, the time- and volume-averaged drop phase velocity turnes out to be larger than for the whole mixture (which is an effect of a finite drop-to-particle size ratio), except for conditions very close to critical where the drops are blocked but the continuous phase still has some room to flow. Our first multidrop-multiparticle simulations for emulsion squeezing through a random granular material confirm this behavior. These results were presented at MFGM2006 international and AIChE 2006 meetings. In collaboration with a graduate student (A. Griggs), we finished the study of an immisible deformable drop motion between two plane walls and are working on the problem of deformable drop sliding down an inclined wall. The steady-state drop velocity is a non-monotonic function of the Bond number (which is most pronounced at 30 deg. wall inclination angle) reaching a maximum at intermediate deformations. The drop-wall spacing is, unexpectedly, a decreasing function of drop-to-medium viscosity ratio. The gap profile is substantially non-uniform ; the steady shapes exhibit neither a discernible "flat spot" nor a simple "pancake" assumed in prior approximate asymptotic theories. Near-contact drop-wall interactions require high surface resolution achieved through multipole acceleration; we currently have this acceleration implemented for the case of matching viscosities. With the help of a summer student, we performed experiments to generally confirm our theoretical calculations, in particular, non-monotonic behavior of the drop velocity.
Bibliography Type:	Description: (Last Updated: 01/10/2008)
Articles in Peer-reviewed Journals	Zinchenko AZ, Davis RH. "A boundary-integral study of a drop squeezing through interparticle constrictions." J Fluid Mech. 2006;564:227-66. http://dx.doi.org/10.1017/S0022112006001479 , Mar-2006
Articles in Peer-reviewed Journals	Griggs AJ, Zinchenko AZ, Davis RH. "Low-Reynolds-number motion of a deformable drop between two parallel plane walls." Int J Multiphase Flow. In Press, Corrected Proof, Available online 27 October 2006. http://dx.doi.org/10.1016/j.ijmultiphaseflow.2006.06.012 , Oct-2006
Articles in Peer-reviewed Journals	Rother MA, Zinchenko AZ, Davis RH. "Surfactant effects on buoyancy-driven viscous interactions of deformable drops." Colloids and Surfaces A: Physicochemical and Engineering Aspects 2006 Jul 20;282-283:50-60. http://dx.doi.org/10.1016/j.colsurfa.2005.11.086 , Jul-2006